



King Abdulaziz University

Faculty of Computing & Information Technology

Computer Science Department

SOLUTION

Mid Term (April 08, 2026)

Course: CPCS-222 (Discrete Structure I)

Term: Spring 2026 (Semester 2)

Points: 60

Time: 80 minutes

Student ID:

Student Name:

***Section:

Question#	CLO/SO	Max. Marks	Obt. marks
1	1/1	10	
2	1/1	10	
3	1/1	10	
4	1/1	10	
5	2/1	10	
6	3/1	10	
TOTAL		60	

Instructor:

Dr. Muhammad Umair Ramzan

Question No. 01 (10 pts)**CLO 1 / SO-1**

Read the following passage carefully and identify five (05) sentences which are “not a proposition”.

On a pleasant morning, Ahmed arrived at the university campus before his class began. The university opens its gates at 8:00 AM every day. He checked his watch and noticed that it was 7:50 AM. As he walked toward the building, he saw that the library was already open. Please close the library door behind you. Ahmed remembered that 10 is greater than 5. He also recalled that the sum of two odd numbers is always even. While thinking, he wrote down the expression $x+2=7$ in his notebook. His friend later mentioned that the Earth revolves around the Sun. What time does the lecture begin today? Ahmed knew that water boils at 100 degrees Celsius under standard conditions. If it rains, then the ground becomes wet. He also wrote the inequality $y>3$ as another example. Wow, the weather is very pleasant today. Finally, he concluded that 15 is divisible by 3.

SOLUTION:

S. No.	Sentence
1	Please close the library door behind you.
2	$x+2=7$
3	What time does the lecture begin today?
4	$y > 3$
5	Wow, the weather is very pleasant today.

Question No. 02 (10 pts)

CLO 1 / SO-1

For the given the logical expression: $(p \oplus q) \wedge (\neg r \vee p)$ construct the truth table.

p	q	r	$p \oplus q$	$\neg r$	$\neg r \vee p$	$(p \oplus q) \wedge (\neg r \vee p)$
T	T	T	F	F	T	F
T	T	F	F	T	T	F
T	F	T	T	F	T	T
T	F	F	T	T	T	T
F	T	T	T	F	F	F
F	T	F	T	T	T	T
F	F	T	F	F	F	F
F	F	F	F	T	T	F

Question No. 03 (10 pts)

CLO 1 / SO-1

For the given the statement, provide the “Inverse”, “Converse” and “Contrapositive” in English

Sentence	<i>Having a valid digital signature is necessary for the approval of the encrypted research grant submission</i>
Inverse	If the encrypted research grant submission is not approved, then it does not have a valid digital signature
Converse	If a submission has a valid digital signature, then the encrypted research grant submission is approved
Contrapositive	If a submission does not have a valid digital signature, then the encrypted research grant submission is not approved

Question No. 04 (10 pts)**CLO 1 / SO-1**

Suppose we have the following propositions:

- p : The system is secure
- q : The user is authenticated
- r : The data is encrypted
- s : Access is granted

Translate the given sentences into propositional logic.

Sentences	Propositional Logic
<i>Access is granted only if the user is authenticated and the data is encrypted</i>	$S \rightarrow (q \wedge r)$
<i>The system is secure if and only if either the data is encrypted or the user is authenticated, but not both</i>	$P \leftrightarrow (q \oplus r)$
<i>If the system is not secure or the user is not authenticated, then access is not granted</i>	$(\neg p \vee \neg q) \rightarrow \neg s$
<i>Access is granted if the system is secure and either the user is authenticated or the data is not encrypted</i>	$(p \wedge (q \vee \neg r)) \rightarrow s$

Question No. 5 (10 pts)**CLO 2 / SO-1**

Show that the following propositional formula is a **tautology**: using Logical Equivalencies

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

Expression	Logical Equivalence Law
$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Given
$\neg[(p \rightarrow q) \wedge (q \rightarrow r)] \vee (p \rightarrow r)$	Implication Law
$\neg[(p \rightarrow q) \wedge (q \rightarrow r)] \vee (\neg p \vee r)$	Implication Law
$\neg(p \rightarrow q) \vee \neg(q \rightarrow r) \vee (\neg p \vee r)$	De Morgan's Law
$\neg(\neg p \vee q) \vee \neg(q \rightarrow r) \vee (\neg p \vee r)$	Implication Law
$\neg(\neg p \vee q) \vee \neg(\neg q \vee r) \vee (\neg p \vee r)$	Implication Law
$(p \wedge \neg q) \vee \neg(\neg q \vee r) \vee (\neg p \vee r)$	De Morgan's Law
$(p \wedge \neg q) \vee (q \wedge \neg r) \vee (\neg p \vee r)$	De Morgan's Law
$(\neg p \vee r) \vee (p \wedge \neg q) \vee (q \wedge \neg r)$	Commutative Law
$[(\neg p \vee r) \vee p] \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Distributive Law
$[\neg p \vee (r \vee p)] \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Associative Law
$[\neg p \vee (p \vee r)] \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Commutative Law
$[(\neg p \vee p) \vee r] \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Associative Law
$(T \vee r) \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Negation Law
$T \wedge [(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Domination Law
$[(\neg p \vee r) \vee \neg q] \vee (q \wedge \neg r)$	Identity Law
$(\neg p \vee r \vee \neg q) \vee (q \wedge \neg r)$	Associative Law
$[(\neg p \vee r \vee \neg q) \vee q] \wedge [(\neg p \vee r \vee \neg q) \vee \neg r]$	Distributive Law
$[\neg p \vee r \vee (\neg q \vee q)] \wedge [(\neg p \vee \neg q) \vee (r \vee \neg r)]$	Associative Law
$[\neg p \vee r \vee T] \wedge [(\neg p \vee \neg q) \vee (r \vee \neg r)]$	Negation Law
$[\neg p \vee r \vee T] \wedge [(\neg p \vee \neg q) \vee T]$	Negation Law
$T \wedge [(\neg p \vee \neg q) \vee T]$	Domination Law
$T \wedge T$	Domination Law
T	Identity Law

Translate the given predicate logic expressions to English sentences. The propositional functions are self-explanatory.

Predicate Logic to English Sentences
$\forall x (Student(x) \rightarrow Studies(x))$
<i>Every student study</i>
$\exists x (Teacher(x) \wedge \neg Experienced(x))$
<i>There exists a teacher who is not experienced</i>
$\forall x (Course(x) \rightarrow \exists y (Student(y) \wedge Enrolled(y, x)))$
<i>Every course has at least one student enrolled in it</i>
$\exists x (Book(x) \wedge \forall y (Student(y) \rightarrow Reads(y, x)))$
<i>There exists a book that every student reads</i>
$\forall x \exists y (Person(x) \rightarrow (Person(y) \wedge Helps(y, x)))$
<i>For every person, there exists another person who helps them</i>

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 7 Logical Equivalences Involving Conditional Statements.

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

$$p \leftrightarrow q \equiv \neg(p \oplus q)$$

$$p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$$

$$p \oplus q \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$$